

Algebraic Specification of Card-Based Cryptographic Protocols in Rocq

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Abstract. Card-based protocols use physical cards to compute with information-theoretic security. I define a Rocq framework with a finite group, its permutation action, and its shuffle distribution as explicit parameters. The framework connects group actions with executed traces and word-shuffle mixing. Its main instance uses the action of $\text{PGL}(2, 7)$ on eight projective points. Under the uniform shuffle distribution, this instance has exact correctness and the recovery ramp $(t, r, n) = (3, 7, 8)$, where t is the privacy threshold, r is the recovery threshold, and n is the number of card positions. It also has perfect view privacy for coalitions of at most three players. Interpreter soundness transfers this privacy to executed traces. These privacy results assume a uniform Boolean secret, a fixed encoded representative, and passive adversaries. The verifier learns the secret. Active security, security after the reveal, and composition are outside the model. No single physical move samples a uniform group element. I therefore analyze a finite implementation that draws a uniform word of length 200 over five generator letters. A checked fiber-counting certificate counts the words that evaluate to each group element. It proves that the resulting word distribution is within L_1 distance 2^{-40} of the uniform group distribution. The theorem transfers to single-card marginals and to a product with any secret prior. Transport theorems then bound approximate coalition-view and executed-trace privacy by 2^{-39} under the word distribution. A five-card family and two related instances with spectral mixing proofs show which algebraic, operational, privacy, and mixing arguments the framework reuses. The two spectral instances import an analytic gap premise, an external assumption used only for their mixing proofs, as their only assumption beyond the classical principles used throughout.

1 Introduction

Card-based cryptography uses face-down physical cards to compute functions without computational assumptions. A shuffle hides a permutation. A reveal exposes selected card faces. Information-theoretic security requires the resulting observation distribution to hide the protected secret. Den Boer’s five-card protocol computes Boolean conjunction and began this line of work [8]. Later protocols used fewer cards and supported more classes of computations [22,20,17].

Mechanized analysis must connect physical operations with their probability distributions. Koch, Schrempf, and Kirsten use bounded model checking to

search finite protocol spaces and check lower bounds [15]. This method checks a bounded state space, meaning one finite collection of protocol states. The target class consists of t -transitive group actions. A bound for one such space does not establish a result for all such actions. The Rocq formalization¹ instead uses a proof assistant whose generic theorems quantify over the group, action, and shuffle distribution. It defines reusable dealer distributions for sampling encoded initial decks, participant views, and executed traces. The resulting group-parametric model proves perfect trace privacy. It also checks a word-shuffle mixing bound that connects a finite generator process with the ideal shuffle law.

The analysis needs two shuffle distributions because the ideal security proof and the finite implementation use different laws. The uniform distribution on the shuffle group is the ideal law for the security proofs. No dealer samples a uniform group element in one physical move. Instead, the dealer repeatedly draws elementary moves from a small generator alphabet. These draws define the word distribution over group generators. The physical status of the generator letters depends on the instance. In the five-card family, the letters are physical cuts. The $\text{PGL}(2, 7)$ letters are group generators, and I do not claim a physical realization for them. The group parameters connect the two analyses. Transitivity of the action gives coalition privacy under the ideal law. By contrast, a small generating alphabet gives a finite executable shuffle whose word distribution can approximate the ideal law. I first prove security in the uniform-shuffle model of Section 2. I then prove that the word distribution approximates the uniform law.

I contribute a group-parametric framework. Its primary $\text{PGL}(2, 7)$ instance has two machine-checked security results.

Theorem A (informal). *For every group element, the $\text{PGL}(2, 7)$ instance recovers the secret from all eight endpoints.² Under the uniform shuffle distribution, a uniform Boolean secret, and the fixed representative dealer that always encodes each secret with its chosen deck, the instance has recovery parameters $(t, r, n) = (3, 7, 8)$. It also has perfect view and executed-trace privacy against every passive coalition of at most three players.*

Theorem B (informal). *The 200-letter word shuffle over five generator letters is within L_1 distance 2^{-40} of the uniform shuffle distribution. The fixed representative dealer uses one chosen deck for each secret. For every passive coalition of at most three players, the two secret-conditioned view distributions are within L_1 distance 2^{-39} of each other. The corresponding executed coalition-trace distributions satisfy the same bound. Executed correctness holds for every word.³*

¹ The companion artifact contains the source, theorem index, assumption report, and build instructions.

² Formal statement at the end of Section 6. The constituent formalized results are `pgl27_run_recovers`, `pgl27_reveal_ambiguous`, `pgl27_seven_reveal_class`, `pgl27_view_leak_k4`, `pgl27_view_indep`, and `pgl27_coalition_trace_secrecy` in `pgg-smc/instances/pgl27`.

I make five contributions.

- I give a common Rocq interface for a finite group, its permutation action, the shuffle distribution, reconstruction, participant views of the cards held by a chosen group of players, and executed traces.
- I verify the den Boer [8] and Kim and Çetinkaya [14] five-card protocols in Rocq as one profile at bias ε , in the sense of Section 3. A profile packages one instance’s data and proofs. I prove executed-run correctness for the conjunction output, executed-trace privacy against one corrupted player, and input hiding given the announced output.
- I prove the exact mutual-information leakage for every reveal pattern in the five-card family. At bias zero, I recover den Boer’s perfect dealing bound. At bias $1/100$, I kernel-check that after seven cuts every single-card endpoint distribution is within L_1 distance 2^{-40} of uniform.
- For the $\text{PGL}(2, 7)$ instance under the uniform shuffle distribution, I prove correctness and the recovery ramp. The ramp records the privacy threshold, the recovery threshold, and the number of card positions. I also prove coalition-view privacy and executed-trace privacy.
- I prove an L_1 bound between a 200-letter word distribution and the uniform group distribution.⁴ I also transfer this bound to the distribution of the card at each endpoint and to product distributions with any secret prior.

Section 2 fixes the protocol flow, the two distributions, and the security scope. Section 3 presents the framework components used in the proofs. Section 4 instantiates these components for the five-card family, whose small deck permits direct case analysis. Sections 5 and 6 construct the $\text{PGL}(2, 7)$ instance. Deck enumeration no longer suffices, so three-transitivity carries the privacy proofs that end in Theorem A. Section 7 proves the word-shuffle results that end in Theorem B. This section is needed because a physical dealer repeats elementary moves instead of sampling one uniform group element. Section 8 compares the additional instances and states the trust base. Section 9 places the results among related work. Section 10 states the remaining extensions and other future directions.

2 Protocol and Security Model

A run uses one secret and a deck of face-down cards. The dealer encodes the secret as a deck arrangement that is valid for the instance. The dealer then

³ Formal statement at the end of Section 7. The constituent formalized results are `pgl27_word_mixing`, `pgl27_word_view_indist`, `pgl27_word_trace_indist`, and `pgl27_word_run_recovers` in `pgg-smc/instances/pgl27`.

⁴ The L_1 distance is $\|P - Q\|_1 = \sum_x |P(x) - Q(x)|$, which is called variation distance in the formal development, with maximum value 2. The common total variation distance is half of it. An L_1 bound of 2^{-40} therefore bounds every observer’s distinguishing advantage by 2^{-41} . The paper’s security number is the one derived from the coalition-view bound: the 2^{-39} view distance of Theorem B bounds every observer’s distinguishing advantage by 2^{-40} . The 2^{-41} figure concerns only the group-element mixing distance.

samples a finite-group shuffle and applies its permutation action to the deck. Each player receives the card at one position. The players reveal their cards to the verifier. The verifier decodes the secret from the revealed arrangement. Each instance section depicts this sequence in a flow diagram.

The fixed secret encoding leaves the shuffle as the only source of randomness. One player observes one card. A coalition view contains the cards at the coalition's positions. The shuffle distributes each small coalition view across many deck arrangements. The privacy theorems state exactly that the resulting coalition-view distribution is independent of the secret. The rest of this section expresses the same flow as probability distributions.

The framework describes a protocol using the following data:

$$(G, \rho, \mu; R), \quad U_G, \quad \mu^{*L}. \quad (1)$$

Here G is a finite shuffle group. The representation $\rho : G \rightarrow S_n$ means that each group element permutes the n card positions. The distribution μ selects one generator letter, which denotes one elementary random move. The real closed field R supplies the probability values. The distribution U_G is uniform on G . I write μ^{*L} for the word distribution at length L . It is the distribution of the group element obtained by evaluating L independent letters.

A dealer samples a secret S and a dealt arrangement D that is valid for the instance. A sampled shuffle g acts on the deck by its permutation action and produces $\rho(g)D$. Position i of the shuffled arrangement holds the card that D places at position $\rho(g)(i)$. Player i observes the view V_i derived from the card at position i . A coalition $C \subseteq \{0, \dots, n-1\}$ observes the following joint view, which collects the observations at its named positions:

$$V_C = (V_i)_{i \in C}. \quad (2)$$

The interpreter runs the dealer, player, and verifier processes. It records every value that a process receives. This record is the player's executed trace T_i . For a coalition C , the coalition trace $T_C = (T_i)_{i \in C}$ collects the traces of its players. The verifier receives all endpoints and applies the decoder, which recovers S . The verifier therefore learns S by design.

The following player process makes the execution model concrete by showing the values that the player receives and reveals⁵:

```

Definition exchange_player (i : 'I_T)
: sproc pgg_dtype data (player_idx i) :=
\pi{ Receive<dealer_idx> #my_hand =>
  Receive<dealer_idx> $shuffle_idx =>
  Reveal<verifier_idx> &(nth ord0 my_hand shuffle_idx) ;
Finish }.
```

In the listing, $\#$ marks a dealt hand, $\$$ marks the public shuffle index, and $\&$ marks a card position. The player receives the hand and shuffle index. The player

⁵ Formalized in `pgg-smc/protocol/card_exchange_pismc.v` as `exchange_dealer`, `exchange_player`, and `exchange_verifier`.

selects one entry and reveals it to the verifier. The verifier receives this entry as its observation. The interpreter records the hand and shuffle index in the player's trace. The trace theorem concerns this process execution. Under the trace-lifting hypothesis, the recorded trace and the static view determine each other. This is the interpreter-soundness property used here. The process therefore reveals no information beyond the static view. Independence of the static view from the secret consequently gives independence of the executed trace from the secret, so the proof reuses view privacy.

The primary execution distribution samples a Boolean secret uniformly. For each secret s , it deals one fixed valid deck D_s that represents the class of s . It then applies an independent uniform shuffle. I call this distribution the fixed representative dealer, and the headline theorems use it. The all-decks dealer instead samples a valid arrangement uniformly from the selected secret class before applying the shuffle. Both dealer distributions use a uniform Boolean prior, so the two secret bits have equal probability.

The security theorems assume passive, honest-but-curious adversaries. A corrupted player observes the protocol but does not deviate from it. The player retains its view and executed trace. Perfect coalition privacy means that S and V_C are independent. I write $S \perp V_C$ for this independence. Perfect trace privacy is stated by the following equality:

$$H(S \mid T_C) = H(S). \quad (3)$$

The protocol protects a dealt secret, as in a secret-sharing scheme [25]. The dealer holds the secret. The players hold card observations that play the role of shares. Privacy limits what coalitions of curious players learn. The protocol gives the players no private inputs. The model does not cover active deviation. It also makes no security claim after the full deck is publicly revealed, and it gives no composition theorem for multiple executions.

The uniform-shuffle model applies one uniform group element. The word-shuffle model instead lets the dealer apply a sequence of elementary moves from a generator alphabet, whose letters name the permitted elementary shuffles. It replaces U_G with μ^{*L} , the distribution obtained by evaluating L independent letters. I compare distributions P and Q using the L_1 distance:

$$\|P - Q\|_1 = \sum_x |P(x) - Q(x)|. \quad (4)$$

The word-shuffle model exposes only the product of the word. The dealer performs 200 physical moves, but no party records an intermediate arrangement. This non-observation rule is a structural assumption of the formalization. The dealer process takes a list of alternative cuts and deals one hand for each cut. Thus a word affects execution only through its evaluated group element. The uniform-shuffle model uses the same observation rule because its single uniform shuffle is also unobserved.

Figure 1 separates the proved paths for the two distributions. The lower path starts from the finite word distribution and its mixing certificate. It yields approximate coalition-view and executed-trace privacy at 2^{-39} .

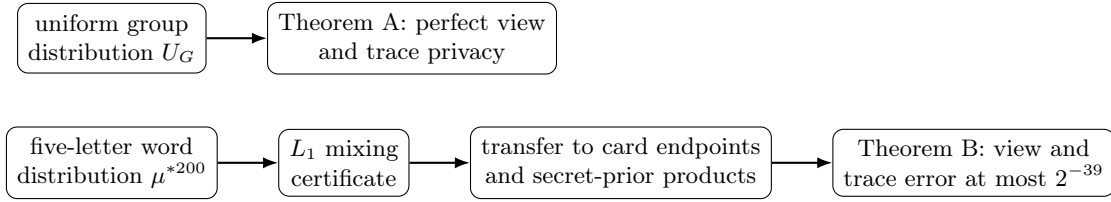


Fig. 1. Two proof paths lead to the headline theorems. The upper path ends at Theorem A. The lower path transfers the mixing certificate from the word distribution to the view and trace bounds of Theorem B.

3 Framework and Generic Theorems

3.1 Framework Architecture

Every instance supplies five kinds of data: a group with its action and generators, a secret type, a run layout, a shuffle-security bound, and a decoder. I gather these choices in one profile record so that one executable protocol can be reused across instances. The generic theorems take their hypotheses directly. I also package the privacy theorem once at the record level. As a result, a profile with a t -transitive action and distinct encoded decks, meaning that no card value is repeated, inherits privacy for coalitions of at most t positions. The profile carries the data of Equation 1 into the executable protocol. The scalar R in Equation 1 is the real closed field, not a protocol datum. The finite algebraic structures use Mathematical Components [30]. Together, these data and obligations give the record an algebraic-specification interpretation. The next paragraph develops that interpretation, and Table 1 then maps each model datum to its record role and to the values supplied by the $\text{PGL}(2, 7)$ instance.

In algebraic specification, the profile record is a parameterized loose specification. Its fields form the signature, which lists the available data and operations. Its stated obligations form the axioms that the data must satisfy. An instance that supplies the fields and proves the obligations is a model of the specification [12,24]. This interpretation explains the five instances in this paper. Each instance is a model of the same specification. It also explains the relation between the ideal and finite shuffle laws. The word-shuffle implementation approximately refines the uniform-shuffle model, and the L_1 bounds of Section 7 quantify the distance. The formal name invokes monodromy because one run transports the deal along a group element. Outside code contexts, I use the term profile record. Because the specification is loose, it also admits models beyond the five in this paper. Natural next instances are the affine group $\text{AGL}(1, 7)$, which maps any ordered pair of distinct positions to any other by exactly one element, and $\text{PGL}(2, 11)$ with the present three-transitive parameters. With this interpretation in place, the following table maps the model data to their record roles.

Model datum	Record role	PGL(2, 7) instance supplies
layout of one protocol run	dealer, player, and verifier processes	eight players, each holding one card
group action (G, ρ) with the generator distribution μ or the uniform group distribution U_G	bound on each card position's endpoint distribution under the shuffle law	three-transitive action and a checked bound for the generator-word distribution
decoder	reconstruction invariant under the group action	orbit-class decoder
run layout, group action and shuffle distribution, and reconstruction	one protocol instance that packages these components	the PGL(2, 7) profile

Table 1. Mapping from model data to framework records. The source names the run-layout interface `PGGInterface`, the shuffle-evidence record `SecurityWitness`, the action-compatible reconstruction record `ReconPlug`, and the aggregate profile `MonodromyProfile`.

The table gives each component's role. Figure 2 shows the dependencies among the components and auxiliary records. It also shows how the profile determines the executable protocol.

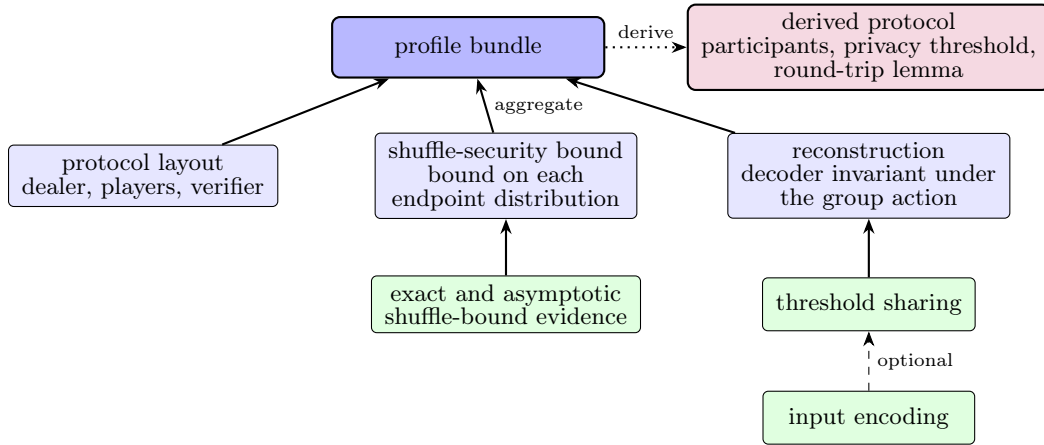


Fig. 2. Architecture of the group-parametric card-protocol framework. Blue boxes denote profile records, green boxes denote auxiliary records, and the purple box denotes the derived protocol. Solid arrows mark required dependencies. The dashed arrow marks an optional dependency. The dotted arrow marks derivation.

Exact slot	Asymptotic slot	Mechanism	Realized by
present	absent	exact equality of the endpoint and target distributions at $\varepsilon = 0$ under the uniform group distribution	den Boer, PGL(2, 7)
present	present	exact finite count giving an error bound that decays geometrically with word length	Kim
absent	present	spectral certificate based on an imported spectral-gap premise	S_5 and the direct product $S_5 \times S_5$

Table 2. Realized combinations of the security witness’s two optional evidence fields. Section 7 gives the word-shuffle counterpart of the PGL(2, 7) row. Section 4 proves the den Boer and Kim rows. Table 4 records the evidence for each instance.

I call a completed profile record a profile. It keeps the participants, verifier, recovery map that decodes the secret, privacy threshold, and shuffle bound tied to one instance. Every profile proves three obligations.

- For every card position, the endpoint-distance field bounds the distance between that position’s post-shuffle card distribution and the uniform distribution by `sw_bound_eps`. The field is `sw_bound`.
- The threshold scheme provides reconstruction and threshold privacy. The field `ts_correct` decodes every valid share tuple to its secret. The field `ts_private` ensures that every coalition below the threshold sees a share pattern compatible with each secret in the same way.
- Reconstruction is invariant under every allowed shuffle. For any valid share tuple, permuting the shares by the shuffle leaves the recovered secret unchanged. The field `rp_recon_invariant` states this property.

The run layout supplies the executable dealer, player, and verifier processes described in Section 2. The group action and shuffle distribution determine the dealing rule and endpoint bound. The reconstruction component selects the decoder. Profiles over the same group can therefore use different reconstruction schemes.

The security witness records the shuffle distribution and its endpoint bound. It may record exact evidence, asymptotic evidence, or both. Table 2 shows the combinations used by the five instances. The den Boer row records only exact evidence because one uniform cut gives the exact endpoint distribution. The PGL(2, 7) row also records only exact evidence. Section 7 proves the word-shuffle bound of Theorem B. This bound is not stored in the record’s optional asymptotic-evidence field. The table therefore reports the record contents, while the evidence for Theorem B remains outside the security witness.

With an input-encoding component, the flow gains a commit prologue. This phase collects the players' inputs and assembles the dealt deck, so the flow computes a function of committed inputs. The realized encoding is den Boer's. Its orbit obligation places assembled layouts for equal-output inputs in one shuffle orbit. Thus, it relates different inputs with the same result. Correctness also requires a valid assembled layout and reconstruction invariant under every allowed shuffle. These facts imply that the shuffled layout reconstructs the intended output. The Kim variant reuses the den Boer program unchanged. With an empty input list, the prologue computes to the plain dealer used for secret sharing. The $\text{PGL}(2, 7)$ instance supplies exactly this empty list. Section 4 realizes the committed-input flow. The instance table in Section 8 gives the full comparison.

Section 4 shows how one instance supplies these components. Once packaged, the reusable security arguments depend on mathematical hypotheses, not on the record representation. The next subsection states these arguments directly.

3.2 Generic Theorems

The first generic theorem derives coalition privacy from transitivity. An instance must provide a t -transitive action on decks whose cards are all distinct. Section 5 proves this requirement for $t = 3$.

Theorem 1 (Generic coalition privacy⁶). *Let G act t -transitively on the card positions. Let S be a Boolean secret with any prior. Suppose that every encoded deck assigns distinct card values to its positions. If g is uniform on G , then*

$$S \perp\!\!\!\perp V_C$$

for every coalition C with $|C| \leq t$.

Theorem 1 is also stated once for every profile. Suppose that the position-permutation group realized by a profile is t -transitive. Suppose also that every encoded deck has distinct cards and that the profile's share count matches its deck length. Then every coalition of at most t positions has a view independent of the secret.⁷ The $\text{PGL}(2, 7)$ instance proves the first two obligations with its three-transitivity and distinct-deck lemmas. Both obligations are necessary for the theorem. An explicit four-position coalition from Section 5 refutes the conclusion when the coalition bound is weakened to $t + 1$. A second profile over the same group deals one constant card to every position. This repeated-card encoder leaks the secret to one position, so it refutes the theorem without the

⁶ Formalized in `pgg-smc/reconstruct/transitivity_privacy.v` as `ttrans_view_indep_gen`.

⁷ `profile_view_indep` in `pgg-smc/reconstruct/transitivity_privacy.v`, discharged by the $\text{PGL}(2, 7)$ instance as `pgl27_view_indep_via_profile` in `pgg-smc/instances/pgl27/pgl27_profile_privacy.v`. The derived corollary restates the instance's view independence verbatim.

distinct-deck obligation.⁸ The record-level statement retains a two-valued secret. The constant-deck counterexample shows that distinct cards cannot be omitted. The mathematics therefore excludes the two-symbol five-card family from this theorem.

Three-transitivity proves privacy for a coalition’s card view. The interpreter instead records an executed trace. The next theorem derives executed-trace privacy when the trace is determined by the view and the view can be recovered from the trace.

Theorem 2 (Generic trace lifting⁹). *Let S , V , and T be finite random variables. Suppose that $T = \tau(V)$. Suppose also that a recovery map ν satisfies $\nu(\tau(v)) = v$ for every v . If V is independent of S , then*

$$H(S \mid T) = H(S). \quad (5)$$

The word-shuffle certificate bounds distributions of group elements. Protocol security instead concerns card positions after the shuffle, coalition views, and traces. The next theorem transfers the bound through any deterministic observable. For a distribution P on A , the pushforward f_*P is the distribution of $f(a)$ when a is sampled from P .

Theorem 3 (Data processing for finite distributions¹⁰). *Let P and Q be finite distributions on A , and let $f : A \rightarrow B$. Their pushforward distributions satisfy*

$$\|f_*P - f_*Q\|_1 \leq \|P - Q\|_1.$$

4 A First Instance: The Five-Card Family

The five-card family is the simplest concrete instance of the framework. One profile covers two classic protocols. The cyclic group C_5 rotates the five-card deck, and the decoder tests whether the three hearts are consecutive. Den Boer’s protocol [8] and the Kim and Çetinkaya variant [14] differ only in their dealing distributions. A bias parameter ε therefore distinguishes them within the same profile. Figure 3 shows one run.

⁸ `profile_view_indep_sharp` and `profile_distinct_deck_necessary` in `pgg-smc/instances/pgl27/pgl27_profile_privacy.v`.

⁹ Formalized in `pgg-smc/security/pgg_trace_secrecy.v` as `trace_secrecy_of_view`.

¹⁰ Formalized in `pgg-smc/security/pgg_collusion_bound.v` as `var_dist_fdmap`.

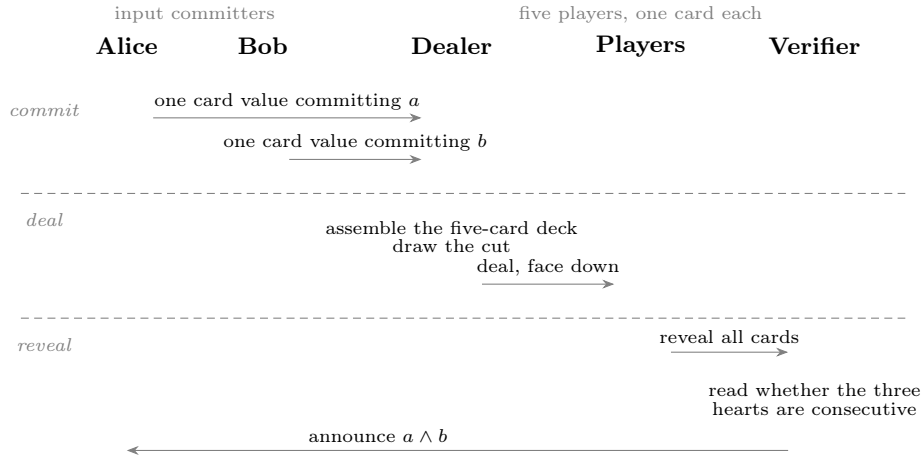


Fig. 3. One five-card run with committed inputs. The physical protocol represents each input bit with two face-down cards. The formalization represents each committed bit with one card value.

The five-card family first shows how one profile combines the framework components. The following definition combines the cyclic action, the security witness for the biased cut, and the three-hearts decoder.¹¹

```

Definition five_card_profile (R : realType) (eps : R)
  (* three bias hypotheses elided *) (L : nat) : MonodromyProfile R :=
  @MkMonodromyProfile R FiveCardKim_M bool FiveCardKim_PI
  (fc_kim_security_witness ... L) (* biased-cut witness *)
  five_card_plug.                (* three-hearts decoder *)
    
```

Kim and Çetinkaya’s biased cut defines the dealing distribution. I denote this distribution by w_ε :

$$w_\varepsilon(a^k) = \begin{cases} \frac{1}{5} - \varepsilon, & k = 0, \\ \frac{1}{5} + \frac{\varepsilon}{4}, & k = 1, 2, 3, 4, \end{cases} \quad -\frac{4}{5} < \varepsilon < \frac{1}{5}. \quad (6)$$

The five weights sum to one. The stated interval makes every weight positive, so w_ε is a valid dealing distribution. At bias zero, the witness bound is zero for every positive word length. In this precise sense, the unbiased member is den Boer’s protocol.¹²

The reveal-set leakage theorem gives the exact leakage for every fixed reveal set.¹³ One revealed card carries no information about the conjunction. This is the

¹¹ Formalized in `pgg-smc/instances/kim2025/five_card_family.v` as `five_card_profile`. The listing elides the three bias hypotheses.

¹² `five_card_eps0_eq0` in `pgg-smc/instances/kim2025/five_card_family.v`.

¹³ The master theorem `leak_view_set` covers all thirty-two position subsets, anchored by `leak_k1`, `leak_k2_adj`, `leak_k2_dist2`, `leak_k3`, `leak_k3_gap`, `leak_k4`, `leak_k5`,

information-theoretic counterpart of the scheme’s privacy threshold. The leakage curve reaches the secret’s entropy, $2 - \frac{3}{4} \log 3 \approx 0.811$. I call this maximum leakage the cap. The reported decimals evaluate the proven closed forms.

The leakage values above concern the announced output. The five-card property in the literature is input-hiding given that output. An observer learns nothing about the input pair beyond the conjunction that the protocol announces. The formalization proves this property for every reveal pattern. Given the announced conjunction, the input pair and the view of any reveal set are conditionally independent, and their conditional mutual information is zero.¹⁴ Conditioning on the announced output is essential. The view contains the announcement, so independence fails without this conditioning.

Figure 4 shows two features of the leakage theorem. For two revealed cards, leakage depends on their relative positions. Four and five revealed cards reach the secret’s entropy.

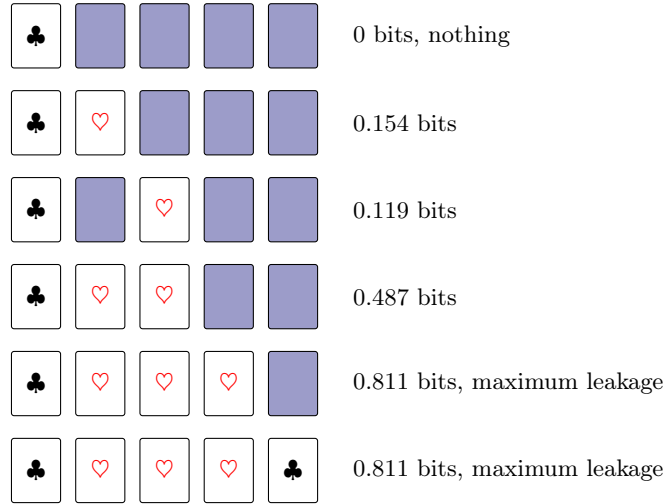


Fig. 4. Mutual-information leakage for selected reveal cases. Blue card backs mark unrevealed positions. Each value depends only on the revealed positions. The displayed faces show one possible outcome and do not determine the value.

Every three-card reveal leaks $\frac{6}{5} - \frac{9}{20} \log 3$ bits. The consecutive and gapped cases therefore have the same value. Only two-card leakage depends on the relative revealed positions.

and the secret’s own entropy H_{secret} in `pgg-smc/instances/denboer1989/five_card_leakage.v`. For example `leak_k2_adj =  $\frac{27}{10} - \frac{1}{4} \log 5 - \frac{7}{10} \log 7$` .

¹⁴ `den_boer_cinde` and `den_boer_input_private` in `pgg-smc/instances/denboer1989/den_boer_encoding.v`, with the reveal-set forms `cinde_ViewS` and `input_private_ViewS`.

Example 1. Let the reveal expose two of the five positions. Two adjacent positions reveal 0.154 bits about the conjunction. Two positions at cyclic distance two reveal 0.119 bits. Both decimals evaluate proven closed forms. In particular, the adjacent value is $\frac{27}{10} - \frac{1}{4} \log 5 - \frac{7}{10} \log 7$. For three revealed cards, the relative positions no longer affect leakage because the closed form depends only on the number of revealed cards.

The example describes the information exposed by a fixed reveal set. The profile also needs a shuffle-security witness for Kim’s repeated biased cut. The next proposition gives this witness at the parameters used here.

Proposition 1 (Seven-cut security at bias $1/100^{15}$). *Let the shuffle distribution be the biased cut $w_{1/100}$ repeated seven times. Then every single-card endpoint distribution is within L_1 distance 2^{-40} of uniform.*

This bound is the profile’s quantitative shuffle witness. It measures how closely one player’s card endpoint follows the uniform-cut ideal after seven biased cuts. The Rocq kernel checks the finite computation.

The two family members share one executed program, so correctness carries over without change. The Kim program is the den Boer program, and its recovery theorem reuses the den Boer proof.¹⁶ The committed inputs of the two players realize the function-evaluation flow of Section 3 with den Boer’s input encoding. Den Boer’s uniform cut gives the exact endpoint distribution after one shuffle. For one corrupted player, the executed trace leaves the secret’s conditional entropy equal to its plain entropy.¹⁷ Table 4 in Section 8 gives the full comparison.

The five-card family uses a cyclic group and a deck small enough for kernel enumeration. The next sections instantiate the same records when enumeration is not feasible. The $\text{PGL}(2, 7)$ construction strengthens coalition privacy beyond one card by using three-transitivity and an orbit-class secret. Its word certificates avoid computation inside the group.

5 The $\text{PGL}(2, 7)$ Construction

I construct the paper’s main eight-player instance in this section. It is an instance of the profile specification, which packages the data and proof obligations for one protocol instance. The construction supplies the record fields of Section 3 and proves their obligations. Each player receives one card as a physical share of the secret. Under the fixed representative dealer and an independent uniform $\text{PGL}(2, 7)$ shuffle, every passive coalition of at most three players has perfect view and executed-trace privacy. Every reveal of seven positions determines the secret class. Every reveal of at most six positions remains compatible with both classes. A view of four positions can leak information, so the privacy cutoff at three is

¹⁵ Formalized in `pgg-smc/instances/kim2025/five_card_kim.v` as `kim_bound_centi` and `kim_deal_centi_lt`.

¹⁶ `kim_procs` and `kim_run_recovers` in `pgg-smc/instances/kim2025/kim_run.v`.

¹⁷ `kim_trace_secrecy` in `pgg-smc/instances/kim2025/kim_trace.v`.

sharp. These thresholds give the recovery ramp $(t, r, n) = (3, 7, 8)$. Decoding the full eight-card arrangement is correct for every allowed group element. Theorem A collects these exact guarantees. Theorem B later replaces the uniform group element with a uniformly sampled 200-letter generator word and bounds the resulting privacy error. To build the instance that supports these results, I first specify its eight card positions.

I index the eight card positions by the projective line over $\mathbb{F}_7$, which consists of seven field elements and one point at infinity:

$$X = \mathbb{P}^1(\mathbb{F}_7) = \{0, 1, \dots, 6, \infty\}.$$

The symbols $0, \dots, 6$ are the seven field elements. The symbol ∞ is the additional projective point. Each point labels one physical position and the player who receives its card. Card values use the separate labels $C = \{0, 1, \dots, 7\}$. Thus all eight cards carry distinct symbols. This standard-deck setting uses pairwise distinguishable cards of the kind found in common decks. Most protocol sizes are instead measured for two-symbol encodings [23,15]. The group $G = \text{PGL}(2, 7)$ acts on X by fractional linear transformations. Each group element therefore permutes the eight card positions and defines an allowed deck rearrangement. The group has order¹⁸:

$$|G| = 336. \tag{7}$$

The formal representation assigns the point at infinity ∞ to position seven.

The $\text{PGL}(2, 7)$ action on X also moves sets of four card positions by moving every position in the set. I encode the secret bit as the orbit class of the four positions occupied by hearts. Every allowed shuffle keeps this set in the same orbit, so the orbit label does not change. This invariance gives correctness under shuffling. Privacy requires a separate argument about the card values that a coalition sees at its named positions.

5.1 Orbit Encoding of the Secret

I make the encoding precise in four steps. First, I define the four-position heart patterns obtained from valid decks under allowed shuffles. Next, I give an explicit classifier for the two orbits. I then choose one deck from each class. Finally, I prove that every allowed shuffle preserves the classifier.

Definition 1 (Orbit of four positions). *Let Ω_4 be the set of all four-position subsets of X :*

$$\Omega_4 = \{A \subseteq X \mid |A| = 4\}.$$

For $A, B \in \Omega_4$, define the orbit and orbit equivalence by

$$\text{Orb}_G(A) = \{g \cdot A \mid g \in G\}, \quad A \sim_G B \iff B \in \text{Orb}_G(A).$$

¹⁸ Formalized in `pgg-smc/instances/pgl27/pgl27_mixing.v` as `pgl27_card`. The order of the abstract quotient is `pgl27_pgl2_order` in `pgg-smc/instances/pgl27/pgl27_group.v`.

Each set $A \in \Omega_4$ records the four positions occupied by hearts but forgets the individual labels of those four hearts. In this subsection, every orbit is an orbit in Ω_4 . Two heart-position patterns are equivalent when an allowed group action carries one to the other. Each valid labelled deck therefore selects one orbit representative, namely its concrete set of heart positions.

To decode an orbit label from a deck, I first extract the heart positions. I then need a test whose value is constant within each orbit and differs between the two orbits. The cross ratio supplies this test.

Definition 2 (Heart-position map and decoder¹⁹). *A valid deck is a bijection $D : X \rightarrow C$. Its four heart-labeled card values are 0, 1, 2, 3. Their positions form*

$$H(D) = \{i \in X \mid D(i) \in \{0, 1, 2, 3\}\} \in \Omega_4.$$

Let $A = \{a < b < c < d\}$ under the order $0 < 1 < \dots < 6 < \infty$. Define the cross ratio in the finite field $\mathbb{F}_7$ by

$$[a, b; c, d] = \frac{(a-c)(b-d)}{(a-d)(b-c)}, \quad [a, b; c, \infty] = \frac{a-c}{b-c}.$$

The second formula covers the projective point at infinity. Because the four points are distinct, their cross ratio is neither zero nor one. The five listed values therefore exhaust the possible elements of $\mathbb{F}_7$. Define the classifier $\kappa : \Omega_4 \rightarrow \{0, 1\}$ by

$$\kappa(A) = \begin{cases} 1, & [a, b; c, d] \in \{3, 5\}, \\ 0, & [a, b; c, d] \in \{2, 4, 6\}. \end{cases}$$

The classifier maps the equianharmonic class to one and the harmonic class to zero. The deck decoder is

$$\text{dec}(D) = \kappa(H(D)), \quad D \xrightarrow{H} \Omega_4 \xrightarrow{\kappa} \{0, 1\}.$$

For κ to be an orbit label, each fiber of κ must equal one orbit of G . A fiber is the set of patterns assigned one classifier value. Thus κ must be constant on each orbit and must give different values to distinct orbits. The next theorem proves both requirements and counts the two fibers.

Theorem 4 (The two orbit classes²⁰). *For all $A, B \in \Omega_4$,*

$$\kappa(A) = \kappa(B) \iff \exists g \in G, B = g \cdot A.$$

Thus the quotient by orbit equivalence has two classes:

$$\Omega_4/G \cong \{0, 1\}, \quad |\kappa^{-1}(0)| = 42, \quad |\kappa^{-1}(1)| = 28.$$

The fiber $\kappa^{-1}(0)$ is harmonic. The fiber $\kappa^{-1}(1)$ is equianharmonic.

¹⁹ Formalized in `pgg-smc/instances/pgl27/pgl27_orbit.v` as `is_heart`, `deck_ok`, `heart_set`, `cross_ratio`, `equianharmonic`, `subset_class`, and `orbit_class`.

²⁰ Formalized in `pgg-smc/instances/pgl27/pgl27_orbit.v` as `subset_class_orbit`, `subset_class_orbitE`, `orbit_class_split`, and `orbit_class_split_complement`.

I use the two stable orbit labels to represent a Boolean secret. For each label $s \in \{0, 1\}$, the dealer fixes one valid representative deck D_s . The dealer later shuffles this deck and gives one card to each player. Orbit invariance will prove correctness. Privacy will follow later from the shuffle distribution and three-transitivity, which makes each ordered observation of at most three positions uniform. The following definition gives the two representative decks and the encoder.

Definition 3 (Orbit encoder²¹). *For each secret $s \in \{0, 1\}$, let D_s be a valid representative deck whose heart-position set has orbit label s . In this instance, take*

$$D_0 = (0, 1, 2, 3, 4, 5, 6, 7), \quad D_1 = (0, 1, 2, 4, 3, 5, 6, 7),$$

and define the encoder by $\text{enc}(s) = D_s$.

Figure 5 shows the different heart-position patterns in the two representative decks.

	0	1	2	3	4	5	6	7
D_0	0	1	2	3	4	5	6	7
D_1	0	1	2	4	3	5	6	7

Fig. 5. The two encoded representatives. Gray labels mark card positions, and position seven represents ∞ . Boxed numbers are card values. Shading marks the four heart-labeled card values. The row D_0 represents the harmonic secret 0. The row D_1 represents the equianharmonic secret 1.

The decoder must assign each representative deck to the bit that its encoder received. The following calculation checks this encoder-decoder round trip for both representatives. Shuffle invariance will then extend the equality to every deck reachable from either representative.

Example 2 (Decoding the representatives²²). The rows in Figure 5 satisfy

$$H(D_0) = \{0, 1, 2, 3\}, \quad \kappa(H(D_0)) = 0,$$

and

$$H(D_1) = \{0, 1, 2, 4\}, \quad \kappa(H(D_1)) = 1.$$

²¹ Formalized in `pgg-smc/instances/pgl27/pgl27_orbit.v` as `orbit_encode` and `orbit_encode_deck`.

²² Formalized in `pgg-smc/instances/pgl27/pgl27_orbit.v` as `orbit_encodeK`.

Thus the complete encoding path applies the encoder, heart-set extraction, and classifier in order:

$$s \mapsto D_s \mapsto H(D_s) \xrightarrow{\kappa} s, \quad \text{dec}(\text{enc}(s)) = s.$$

The example proves $\text{dec}(\text{enc}(s)) = s$ before shuffling. Correctness of a full run also requires the decoder to return the same value after every allowed shuffle. Let $\rho : G \rightarrow S_X$ be the permutation of card positions induced by the group action. A deck $D : X \rightarrow C$ maps each position to its card value. Define the shuffled deck by:

$$(g \star D)(i) = D(\rho(g)(i)).$$

The symbol $\star$ denotes this deck rearrangement. In $g \star D$, position i contains the value that D places at $\rho(g)(i)$. The set of new heart positions therefore pulls the old heart set back through $\rho(g)$, which produces the inverse position action. The following lemma states this heart-set transport.

Lemma 1 (Shuffle invariance²³). *For every $g \in G$ and valid deck D , the heart positions move by the inverse positional action:*

$$H(g \star D) = \rho(g)^{-1} \cdot H(D).$$

The shuffled deck remains valid, and its decoded value does not change:

$$g \star D \text{ is valid}, \quad \text{dec}(g \star D) = \text{dec}(D).$$

The equality $\rho(g)^{-1} = \rho(g^{-1})$ shows that the inverse permutation comes from the inverse group element. Thus the shuffled heart set remains in the same G -orbit as $H(D)$. The shuffle may change the orbit representative in Ω_4 , but the orbit label remains fixed.

In one run, the dealer encodes s as D_s , applies an allowed shuffle, and gives one card to each player. When all eight players reveal their cards, the verifier sees $g \star D_s$. Proposition 2 later lifts deck-level correctness to the executable interpreter. The recovery threshold asks how many revealed positions determine the secret. Coalition privacy asks what smaller named groups learn. Section 6 proves these two separate properties. Those later arguments depend first on deck-level correctness. The following corollary states the equality required by this execution.

Corollary 1 (Shuffle correctness of orbit decoding²⁴). *For every $s \in \{0, 1\}$ and $g \in G$,*

$$\text{dec}(g \star \text{enc}(s)) = \text{dec}(\text{enc}(s)) = s.$$

²³ Formalized in `pgg-smc/instances/pgl27/pgl27_orbit.v` as `subset_class_invariant`, `orbit_class_invariant`, and `deck_stable`. The heart-set identity is the local equality `Hheart` in the proof of `orbit_class_invariant`, not a separate exported theorem.

²⁴ The direct equality is formalized in `pgg-smc/instances/pgl27/pgl27_orbit.v` as `orbit_encodeK` and `orbit_class_invariant`. The reconstruction-interface counterpart is `orbit_recon_invariant` in `pgg-smc/instances/pgl27/pgl27_scheme.v`.

Corollary 1 shows that every allowed shuffle preserves the decoded secret of an encoded deck. Correctness and privacy require separate arguments. Privacy concerns the distribution of partial observations when $g \leftarrow U_G$ is uniform and independent of the secret. A coalition's partial observation contains only the card values at its named positions. The privacy proof uses three-transitivity, which lets the action carry any ordered triple of distinct positions to any other. Before I give that proof, I use the next figure to return from the deck equality to one complete execution.

Figure 6 shows where the encoder and decoder act during one execution.

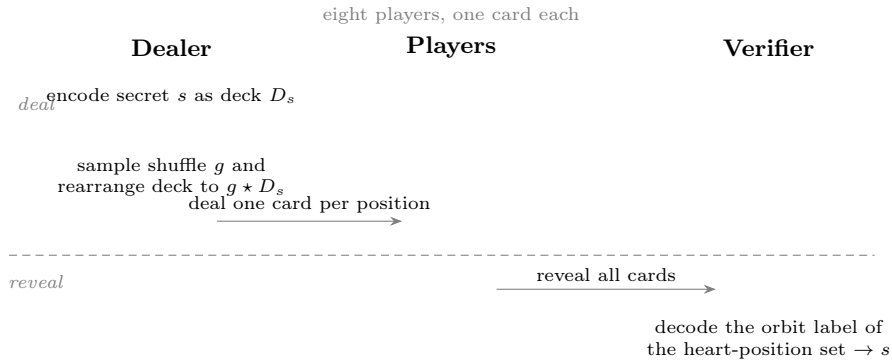


Fig. 6. One $\text{PGL}(2, 7)$ run with eight players and one card per player. The players supply no private inputs. The verifier decodes the orbit label of the heart-position set. In the uniform-shuffle model, g follows the uniform distribution U_G . In the word-shuffle model, g follows the word distribution μ^{*L} . This run is the no-input counterpart of Figure 3.

5.2 Action and Generators

The protocol uses the permutation action $\rho : G \rightarrow S_X$ on the eight card positions. A generator is one element of a fixed set whose finite products reach every element of G . Under ρ , each generator rearranges the eight cards once. A generator word lists a sequence of these elementary rearrangements. Evaluating the word gives its product $g \in G$. In the passive execution model, no party observes an intermediate arrangement. The word therefore affects execution only through g and the final deck $g \star D_s$.

For $G = \text{PGL}(2, 7)$, three fractional linear transformations generate the group. Translation is $z \mapsto z + 1$. Scaling is $z \mapsto 3z$. Inversion is $z \mapsto -1/z$.²⁵ On the projective line, translation and scaling fix ∞ , while inversion exchanges 0 and ∞ . Figure 7 shows each map as a rearrangement of eight cards, one at each projective

²⁵ The rows of Figure 7 are the permutation tables of the three maps, identified with them by `tr_moebius`, `sc_moebius`, and `inv_moebius` in `pgg-smc/instances/pgl27/pgl27_group.v`.

	0	1	2	3	4	5	6	7
$z \mapsto z$	0	1	2	3	4	5	6	∞
$z \mapsto z + 1$	1	2	3	4	5	6	0	∞
$z \mapsto 3z$	0	3	6	2	5	1	4	∞
$z \mapsto -1/z$	∞	6	3	2	5	4	1	0

Fig. 7. The three generating maps on the projective points. Each row shows one elementary group transformation. Gray labels mark positions. The box at position i contains its image point $g(i)$. Position seven represents ∞ .

point. The five-letter alphabet used by the executable shuffle in Section 7 adds the inverses of translation and scaling to these three maps. The group order is $|G| = 336 = 8 \cdot 7 \cdot 6$, which is also the number of ordered triples of distinct points. Lemma 2 provides a group element for each ordered destination triple. The count then makes this element unique.

The figure gives the three permutation tables. The following example first reads each table as a physical movement of the card row. It then shows how a generator word sends one ordered triple of source points to an ordered triple of target points.

Example 3. Start with the identity row, which places label z at position z . Translation changes $(0, 1, 2, 3, 4, 5, 6, \infty)$ to $(1, 2, 3, 4, 5, 6, 0, \infty)$ because position i displays $g(i)$. Scaling fixes 0 and ∞ . It cycles the other six points in one six-cycle. Inversion is an involution, so applying it twice gives the identity. It exchanges 0 with ∞ , 1 with 6, 2 with 3, and 4 with 5. Words in these three maps reach all 336 group elements. Lemma 2 gives one such word for every ordered triple of distinct points. For example, apply inverse scaling, inverse translation, inversion, scaling, and inverse translation in that order. This five-letter generator word evaluates to $z \mapsto (z - 1)/(3 - z)$ and sends $(0, 1, 2)$ to $(2, 0, 1)$.

5.3 Three-Transitivity and Local Views

The action of $\text{PGL}(2, 7)$ on the eight projective points is sharply three-transitive. Thus exactly one group element sends any ordered triple of distinct points to any other. The uniqueness follows because the group order is $336 = 8 \cdot 7 \cdot 6$, which is the number of ordered triples. Under a uniform shuffle, every ordered view of three positions is uniform and carries no group invariant. A view of four positions carries one invariant, the cross ratio. This algebraic fact has four consequences.

It makes the cross-ratio decoder possible. It gives perfect privacy to coalitions of at most three players. It forces coalitions of four players to leak information. It also explains the choice of $\text{PGL}(2, 7)$ for the construction.

Global orbit invariance proves correctness. Privacy asks what card values a fixed coalition sees when $g \leftarrow U_G$ is independent and uniform. The coalition observes named positions, so an ordered tuple must preserve which player saw each value.

Lemma 2 (Three-transitivity²⁶). *For any ordered triples (x_1, x_2, x_3) and (y_1, y_2, y_3) of distinct projective points, there exists $g \in \text{PGL}(2, 7)$ such that*

$$g(x_1) = y_1, \quad g(x_2) = y_2, \quad g(x_3) = y_3.$$

The formal proof enumerates generator words from one fixed base triple to every distinct target triple. The Rocq kernel checks the resulting finite table. The base triple is $(0, 1, 2)$. For every other ordered triple of distinct points, the table stores one generator word that sends $(0, 1, 2)$ to that triple. Lemma 2 then makes every ordered view of at most three positions uniform under U_G . This lemma supplies the group-action premise of Theorem 1 at $t = 3$. The generic theorem gives uniform-shuffle coalition privacy. Section 7 treats the nonuniform word distribution separately because it requires a mixing argument.

Table 3 at the end of Section 7 lists the formal source locations for this section and the next two sections.

6 Correctness, Recovery, and Uniform-Shuffle Privacy

The construction section proved deck-level correctness and the group-action condition needed for privacy. This section lifts correctness from decks to full protocol executions. It also determines the recovery ramp and proves view and trace privacy under a uniform shuffle. Together, these results form Theorem A.

6.1 Correctness and Recovery

The interpreter deals D_s , applies an allowed shuffle, and sends one card to each of eight players.²⁷ The players return their cards to the verifier. The verifier's endpoint sequence is the shuffled dealt arrangement. Orbit-class invariance and the encoder equation, which identifies the decoded orbit class with the encoded secret, then prove correctness.

Proposition 2 (Executed correctness²⁸).

For every $s \in \{0, 1\}$ and $g \in \text{PGL}(2, 7)$, decoding all eight verifier endpoints after the shuffle g returns s .

²⁶ Formalized in `pgg-smc/instances/pgl27/pgl27_group.v` as `pgl27_3transitive`.

²⁷ The dealer program is `exchange_dealer` and the $\text{PGL}(2, 7)$ witness is the singleton cut list in `pgl27_dealer_run`.

²⁸ Formalized in `pgg-smc/instances/pgl27/pgl27_run.v` as `pgl27_run_recovers`. The direct class-recovery counterpart is formalized in `pgg-smc/instances/pgl27/pgl27_trace.v` as `pgl27_run_recovers_class`.

Because correctness holds for every group element, it holds with probability one under both the uniform distribution and any word distribution. In particular, the executed run is correct for the evaluated 200-letter word.²⁹ By contrast, only privacy incurs an approximation error. Having settled full-run correctness for both shuffle laws, I next ask how many revealed positions suffice to determine the secret.

The structural recovery threshold is finer than the implemented decoder interface. Any seven revealed positions determine the missing card because every valid deck uses each of the eight card values exactly once. The seven cards therefore determine the orbit class. By contrast, any set of at most six revealed positions is compatible with valid arrangements from both secret classes.

I summarize the instance by the triple (t, r, n) . Its first two parameters answer different questions. The privacy threshold t asks how large a coalition can be while its view remains independent of the secret. It is the largest coalition size with perfect privacy, so it is a probabilistic condition. By contrast, the recovery threshold r asks how many revealed positions always determine the secret class. It is the least such number, so it is a possibilistic condition. Below r , both secret classes remain consistent with the revealed cards. Finally, n is the number of card positions. Following ramp schemes in the secret-sharing literature, I call (t, r, n) the recovery ramp [7,32]. The gap between t and r is the graded regime between privacy and recovery. Although $r = 7$, the implemented decoder reads all eight endpoints. Thus, $r = 7$ is a structural property of the arrangements. It is not the reading rule of a smaller implemented decoder.

Proposition 3 (Recovery ramp and sharpness³⁰). *Every reveal set of at most six positions is compatible with valid arrangements from both secret classes. Every set of seven revealed positions determines the secret class. For the identity arrangement, the coalition consisting of the four heart positions has a view that depends on the secret and has positive mutual information with it.*

Recovery and privacy have different cutoffs. Perfect coalition privacy holds for coalitions of at most three positions. Deterministic recovery starts with seven revealed positions. Figure 8 shows these cutoffs on separate tracks.

The witness coalition shows that the privacy cutoff of three is sharp. Leakage for reveal sets of sizes four through six reduces to four values, one for each orbit under G . Four-position reveal sets form two orbits of sizes 42 and 28, separated by the classical cross-ratio invariant. Five-position reveal sets form one orbit, and six-position reveal sets form one orbit.³¹

²⁹ The corollary is `pgl27_word_run_recovers` in `pgg-smc/instances/pgl27/pgl27_word_privacy.v`.

³⁰ Formalized in `pgg-smc/instances/pgl27/pgl27_recovery.v` as `pgl27_reveal_ambiguous` and `pgl27_seven_reveal_class`, and in `pgg-smc/instances/pgl27/pgl27_secrecy.v` as `pgl27_view_dep_k4` and `pgl27_view_leak_k4`.

³¹ The orbit census and the collision counts are kernel-checked in `pgg-smc/instances/pgl27/pgl27_leakage_census.v`, as `pgl27_orbit_four_cover`,

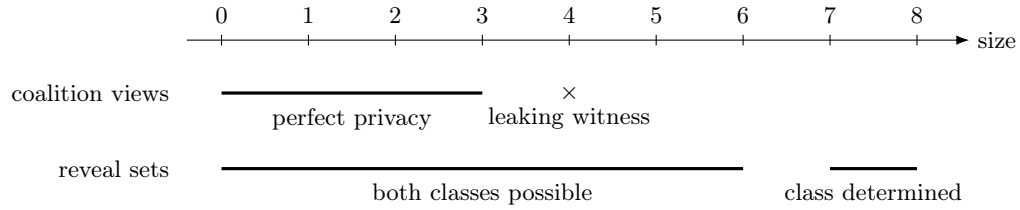


Fig. 8. The recovery ramp $(t, r, n) = (3, 7, 8)$. The view track shows perfect privacy for coalitions of at most three positions and leakage for the four-position witness. The reveal track shows ambiguity through six positions and determination from seven.

Fix an orbit representative C with at least three positions. For either secret value, restriction of the two deals to C is injective. The coalition’s observation is therefore uniform over 336 equally likely possibilities. Let m be the number of observations shared by the two deals. A shared observation is compatible with both secrets, while every other observation identifies one secret. Under the uniform secret prior, the nonshared fraction is $1 - m/336$. This is the mutual information with the secret class in bits. Sharp three-transitivity justifies this identity by a pen-and-paper argument. The identity does not apply below three positions, where the true leakage is zero. The kernel-checked collision counts for the four representatives are $m = 96, 72, 36$, and 12 . The corresponding leakage values are $5/7, 11/14, 25/28$, and $27/28$ bits. The committed script `pgg-smc/instances/pgl27/pgl_leakage_targets.py` evaluates these values from the counts outside the kernel. The four-heart witness represents the orbit of size 42 and has leakage exactly $5/7$ bits. A pen-and-paper relabeling argument extends each value from its representative to the full orbit because a group element preserves leakage. The formalization does not include this relabeling-invariance argument.

6.2 Perfect View and Trace Privacy

The main uniform-shuffle execution samples S uniformly from $\{0, 1\}$. It deals the fixed representative D_S , meaning the chosen encoded deck for secret S .³² It then samples an independent shuffle $g \leftarrow U_G$. Three-transitivity means that any ordered triple of distinct positions can be moved to any other. This property makes the view independent of the secret for every coalition C with $|C| \leq 3$.

`pgl27_five_subset_orbitE`, `pgl27_six_subset_orbitE`, and the `pgl27_collisions` lemmas.

³² The execution distribution is `pgl27P` and the representative is the formal orbit encoder `orbit_encode`.

Proposition 4 (Perfect privacy for the fixed dealer³³). *Assume the uniform Boolean prior, the fixed representative dealer that uses one chosen deck for each secret, and an independent uniform $\text{PGL}(2, 7)$ shuffle. Then*

$$S \perp V_C \quad \text{and} \quad H(S \mid T_C) = H(S) \quad \text{for every } |C| \leq 3.$$

Example 4. Let the dealt secret be $s = 0$, so the dealt arrangement is D_0 . Let the shuffle be the translation $x \mapsto x + 1$, which leaves the point ∞ fixed at position seven. Each position then shows the card that D_0 places one position later. The coalition $\{0, 1, 2\}$ sees the values $(1, 2, 3)$. For $s = 1$, the same shuffle would show $(1, 2, 4)$. By Lemma 2, another shuffle sends the coalition's positions to positions 1, 2, and 4 of D_1 . These positions contain the values 1, 2, and 3. Thus, three observed cards never distinguish the two secrets. The preceding proposition makes this equality of view distributions exact.

Two additional dealers test the robustness of the fixed representative choice. The all-decks dealer checks whether that choice is special. It samples S uniformly. Given $S = s$, it samples a valid arrangement uniformly from class s . It then applies an independent uniform $\text{PGL}(2, 7)$ shuffle.³⁴

Proposition 5 (All-decks perfect privacy³⁵). *Assume the uniform Boolean prior, a dealer that samples a deck uniformly from the chosen secret class, and an independent uniform $\text{PGL}(2, 7)$ shuffle. Then*

$$S \perp V_C \quad \text{and} \quad H(S \mid T_C) = H(S) \quad \text{for every } |C| \leq 3.$$

A third dealer samples a valid deck uniformly from the chosen secret class and applies no later shuffle. I call this law the shuffle-free deck distribution. Under this law, view independence holds for every Boolean secret prior. The available executed-trace theorem assumes the uniform Boolean prior. I do not claim a prior-generic trace theorem.

Proposition 6 (Shuffle-free deck privacy³⁶). *For any Boolean prior, a deck sampled uniformly from the chosen secret class gives $S \perp V_C$ whenever $|C| \leq 3$. Thus, the coalition view is independent of the secret. Under the uniform Boolean prior, the same dealer also gives $H(S \mid T_C) = H(S)$, so the executed trace preserves the secret's entropy.*

³³ Formalized in `pgg-smc/instances/pgl27/pgl27_secret.v` as `pgl27_view_indep`. The trace counterpart is formalized in `pgg-smc/instances/pgl27/pgl27_trace.v` as `pgl27_coalition_trace_secret`.

³⁴ The all-decks distribution is `pgl27P_alldecks`.

³⁵ Formalized in `pgg-smc/instances/pgl27/pgl27_secret.v` as `pgl27_view_indep_alldecks`. The trace counterpart is formalized in `pgg-smc/instances/pgl27/pgl27_trace.v` as `pgl27_alldecks_coalition_secret`.

³⁶ Formalized in `pgg-smc/instances/pgl27/pgl27_secret.v` as `pgl27_view_indep_deck_prior`. The trace counterpart is formalized in `pgg-smc/instances/pgl27/pgl27_trace.v` as `pgl27_deck_coalition_secret`.

Executed correctness gives decoding for the full run. The recovery-ramp proposition gives the recovery threshold and the four-card leakage witness. Fixed-dealer privacy gives view and trace privacy through coalition size three. This makes the four-card witness sharp. Together, these results prove Theorem A.

Theorem A. *For every group element, the $\text{PGL}(2, 7)$ instance recovers the secret from all eight endpoints.³⁷ Under the uniform Boolean prior, the fixed representative dealer that uses one chosen deck for each secret, and an independent uniform $\text{PGL}(2, 7)$ shuffle, the instance has recovery parameters*

$$(t, r, n) = (3, 7, 8),$$

and for every passive coalition C with $|C| \leq 3$,

$$S \perp V_C \quad \text{and} \quad H(S \mid T_C) = H(S).$$

Section 8 states the trust base for Theorem A.

Theorem A assumes a uniform group element. The next section replaces this ideal shuffle with a finite generator word and quantifies the resulting error.

7 Word-Shuffle Approximation of the Uniform-Shuffle Model

Section 2 defines generator words. This section proves a machine-checked mixing certificate for the finitely generated shuffle. It shows that evaluating a uniform word gives a distribution close to the uniform distribution on the group. The letters are group generators, not cuts of the eight-card row. I do not claim a physical realization of an unobserved letter application. The main certificate bounds the distance between the executable and ideal shuffles. The proof then transfers this bound to security observables. Theorem B states the final bound.

The executable shuffle samples a word over five letters. Three letters are the projective translation, scaling, and inversion generators. The other two are the inverses of translation and scaling. Inversion is self-inverse. The fiber computation steps backward through each letter's inverse, so every inverse must remain in the alphabet. This symmetric five-letter generator tuple generates the same group as the original three generators. The fiber count therefore still concerns the same 336 group elements.

For word length L , the shuffle distribution evaluates a uniform word in $\{0, 1, 2, 3, 4\}^L$. At the length $L = 200$ used in Theorem B, the sample space contains 5^{200} words. The proof partitions these words by their evaluated group

³⁷ This statement is the conjunction of the following formalized results: `pgl27_run_recovers` in `pgg-smc/instances/pgl27/pgl27_run.v`, `pgl27_reveal_ambiguous` and `pgl27_seven_reveal_class` in `pgg-smc/instances/pgl27/pgl27_recovery.v`, `pgl27_view_leak_k4` and `pgl27_view_indep` in `pgg-smc/instances/pgl27/pgl27_secrecy.v`, and `pgl27_coalition_trace_secrecy` in `pgg-smc/instances/pgl27/pgl27_trace.v`.

element. I call the words that evaluate to g the fiber of g . The 336 group elements therefore define 336 fibers. Informally, the certificate counts how often each group element occurs among the 5^{200} words and checks its deviation from the average. For example, at $L = 1$, the translation letter evaluates to $z \mapsto z + 1$ and belongs to that group element's fiber. A computation using binary natural numbers records every fiber size. The final arithmetic check sums the deviations from the average size $5^{200}/336$.

Lemma 3 (Word-shuffle mixing bound³⁸). *Let μ be uniform on the five-letter symmetric generator tuple. Then*

$$\|\mu^{*200} - U_G\|_1 \leq 2^{-40}. \quad (8)$$

Equation 8 is a checked mixing certificate. Its inputs are the five-letter generation theorem, the group order in Equation 7, the 336 fiber counts, and the final integer inequality. The certificate concerns only the distribution of shuffle group elements.

A coalition observes card endpoints, not group elements. The proof therefore transfers the bound from group elements to endpoints. The endpoint transfer maps each permutation to the endpoint of one fixed card. Here $h_{\#}P$ denotes the distribution of $h(g)$ when g is sampled from P .

Lemma 4 (Endpoint transfer³⁹). *By the L_1 data-processing inequality, for every card position i ,*

$$\|(g \mapsto \rho(g)i)_{\#}\mu^{*200} - (g \mapsto \rho(g)i)_{\#}U_G\|_1 \leq 2^{-40}. \quad (9)$$

The endpoint transfer controls one card position. Transitivity makes the endpoint distribution under the uniform shuffle uniform on the eight positions. The privacy theorem also includes the secret variable. The next lemma forms the product of each shuffle distribution with the same independent Boolean prior and preserves the L_1 bound.

Lemma 5 (Secret-prior product transfer⁴⁰). *Let π be any distribution on Boolean secrets. The product distributions make the secret and shuffle independent, and*

$$\|\pi \times \mu^{*200} - \pi \times U_G\|_1 \leq 2^{-40}. \quad (10)$$

Equations 9 and 10 apply the single certificate in Equation 8 to two additional sample spaces. Neither equation is a separate mixing certificate. Equation 10 compares a secret prior paired with a shuffle distribution. It does not yet describe executed runs.

The proof chain establishes proximity of the group distributions, every single-card marginal, and the product with any secret prior. Transport through the protocol interpreter then turns the certificate into Theorem B.

³⁸ Formalized in `pgg-smc/instances/pgl27/pgl27_mixing.v` as `pgl27_word_mixing`.

³⁹ Formalized in `pgg-smc/instances/pgl27/pgl27_mixing.v` as `pgl27_endpoint_mixing`.

⁴⁰ Formalized in `pgg-smc/instances/pgl27/pgl27_mixing.v` as `pgl27_joint_mixing`.

Theorem B. *Let μ be uniform on the five-letter symmetric generator tuple.⁴¹*

Then

$$\|\mu^{*200} - U_G\|_1 \leq 2^{-40}.$$

For the fixed representative dealer, every passive coalition C with $|C| \leq 3$, and all secrets $s, s' \in \{0, 1\}$, let $V_C(s, g)$ denote the coalition view at dealt secret s and shuffle g . Then

$$\|(g \mapsto V_C(s, g))_{\#}\mu^{*200} - (g \mapsto V_C(s', g))_{\#}\mu^{*200}\|_1 \leq 2^{-39}, \quad (11)$$

and the executed coalition trace satisfies the same bound. Decoding all eight endpoints returns the dealt secret for every word.

The proof uses a triangle inequality through the uniform group distribution. By three-transitivity, the coalition-view distribution under the uniform shuffle does not depend on the secret. The data-processing inequality applies the certificate in Equation 8 to both sides of the triangle. A companion theorem also gives a bound of 2^{-40} for any Boolean prior. It compares the joint view-and-secret distribution under the word shuffle with the product of its uniform-shuffle marginals.⁴²

The word-shuffle privacy results cover only the fixed representative dealer. The all-decks and shuffle-free dealers use different sample spaces, so the product-distribution transfer does not apply to them. Section 8 states the trust base for these results.

The PGL(2, 7) argument has four proof layers: construction, recovery, privacy, and finite-word transfer. Table 3 maps the mathematical claims used by Theorems A and B to the Rocq results that establish them.

⁴¹ This statement is the conjunction of the following formalized results: `pgl27_word_mixing` in `pogg-smc/instances/pgl27/pgl27_mixing.v`, and `pgl27_word_view_indist`, `pgl27_word_trace_indist`, and `pgl27_word_run_recovers` in `pogg-smc/instances/pgl27/pgl27_word_privacy.v`.

⁴² The companion result is `pgl27_view_mixing`. It is an ideal-proximity statement, not the privacy definition.

Mathematical statement	Rocq source name
<i>Construction</i>	
three-transitivity	pgl27_3transitive
equianharmonic and harmonic or-bit counts	orbit_class_split, orbit_class_split_complement
encoder returns its orbit class	orbit_encodeK
group order	pgl27_card
<i>Correctness and recovery</i>	
executed recovery of the orbit class	pgl27_run_recovers_class
seven cards determine the deck and orbit class	pgl27_seven_reveal_determines, pgl27_seven_reveal_class
at most six revealed cards remain ambiguous	pgl27_six_reveal_ambiguous, pgl27_reveal_ambiguous
four-card view dependence and leakage	pgl27_view_dep_k4, pgl27_view_leak_k4
<i>Privacy</i>	
fixed-representative views	pgl27_view_indep, pgl27_view_leakage_le
fixed-representative traces	pgl27_trace_secrecy, pgl27_coalition_trace_secrecy
all-decks views and traces	pgl27_view_indep_alldecks, pgl27_alldecks_trace_secrecy, pgl27_alldecks_coalition_secrecy
shuffle-free views and traces	pgl27_view_indep_deck_prior, pgl27_deck_trace_secrecy, pgl27_deck_coalition_secrecy
<i>Mixing and transfers</i>	
word mixing certificate	pgl27_word_mixing
endpoint and product transfers	pgl27_endpoint_mixing, pgl27_joint_mixing
word view and trace bounds	pgl27_word_view_indist, pgl27_word_trace_indist
ideal-proximity companion	pgl27_view_mixing
word correctness	pgl27_word_run_recovers

Table 3. Rocq sources for the PGL(2, 7) results.

8 Other Instances and Trust Base

The framework shows which arguments transfer unchanged across instances and which require finite or spectral evidence for a specific instance. In addition to the instances in Sections 4 and 5, the S_5 and $S_5 \times S_5$ instances use the spectral part of the framework. Table 4 records the formally proved result families for all five instances. The trace-lifting theorem of Section 3 is used unchanged by all five instances. Each instance supplies its own endpoint bound. Perfect privacy in the table refers to the dealer distribution defined by that instance. The trace column

Instance	Recovery	Perfect privacy	Trace transport
den Boer	five-card AND	one card	one player
Kim	five-card AND	one card	one player
S_5	additive 5-of-5	fewer than five positions	one player
$S_5 \times S_5$	two additive 5-of-5 shares, each reconstructed from all five positions	fewer than five cards in each five-card pile	one player
PGL(2, 7)	$(t, r, n) = (3, 7, 8)$: privacy threshold, recovery threshold, and player count	view privacy for coalitions of at most three	trace transport for coalitions of at most three
Instance	Word-shuffle mixing	Trust base	
den Boer	exact mixing after one uniform cut	kernel, boolp	
Kim	$L = 7, L_1 < 2^{-40}$	kernel, boolp	
S_5	word length $L = 286$ via a spectral inequality	imported Rayleigh bound for spectral mixing	
$S_5 \times S_5$	within-pile decay, exact global L_1 floor 1 because cards never change from the S_5 instance piles	Rayleigh bound inherited from the S_5 instance	
PGL(2, 7)	$L = 200, L_1 \leq 2^{-40}$	kernel, boolp	

Table 4. Proof coverage across the five protocol instances. The mixing column uses the L_1 distance defined in Equation 4.

does not record a second security property. It records interpreter soundness carrying the same view privacy to the executed traces of the stated corrupted players.

The PGL instance adds coalition views, coalition traces, the recovery ramp, and the finite fiber certificate from Section 7. A coalition trace collects the executed traces of the corrupted players. The finite fiber certificate gives exact evidence for finite-word mixing. This mixing result does not use the imported Rayleigh premise required by the S_5 spectral route.

A result’s trust base consists of its checker and the axioms on which its verification depends. In Table 4, **boolp** is shorthand for the three classical principles from the MathComp probability stack. Propositional extensionality equates logically equivalent propositions. Dependent functional extensionality equates dependent functions that are pointwise equal. Constructive indefinite description selects a witness from a proof that one exists. The S_5 and $S_5 \times S_5$ mixing bounds also import a Rayleigh-quotient premise as an axiom. This analytic premise is used only by the spectral bounds. The PGL(2, 7) group, orbit, cardinality, correctness, and recovery results are proved without additional local

axioms beyond the global assumptions. The Rocq kernel checks every accepted proof term. It also checks the results of finite computations performed by the virtual machine, which evaluates those computations.

The S_5 instance represents its secret by additive sharing modulo five. Its endpoint mixing proof applies a spectral inequality at word length 286. This inequality controls convergence at that length. The development imports the required Rayleigh bound as an axiom. The $S_5 \times S_5$ instance uses two disjoint piles and inherits the same Rayleigh premise. The premise gives decay within each five-card pile. A card never moves between the piles. Its endpoint distribution therefore cannot converge to the uniform distribution on all ten positions, which assigns mass to positions in both piles. The formal global bound approaches the exact L_1 floor of one.

9 Related Work

Den Boer introduced the first card-based cryptographic protocol with the five-card trick [8,16]. It uses five cards to compute the AND of two secret bits with perfect security against passive participants. Later protocols reduced the number of cards and supported other functions [22,20,17].

Koch, Schrempf, and Kirsten use software bounded model checking over encoded finite protocol spaces [15]. Their method searches for protocols and checks finite lower bounds. The Rocq formalization instead proves reusable lemmas for group actions, execution, privacy, and mixing. The Kim instance also connects a biased shuffle model with a concrete word-shuffle bound [14].

Shinagawa specifies a card protocol through a deck and a set of operations in a unified model [26]. This model covers binary, regular polygon, and dihedral cards. Shinagawa varies the card type and allowed physical operations. This paper instead varies the finite group, permutation action, shuffle distribution, and reconstruction map. It connects these parameters with executable traces and machine-checked privacy and mixing results.

Mizuki and Shizuya’s abstract-machine model provides the field’s standard operational semantics [21]. It fixes the sequences of shuffles, turns, and permutations that a protocol may apply. This paper’s interpreter provides the analogous execution model for group-parametric protocols. It draws the shuffle from a word distribution over the group’s generators.

Work on closed shuffles and graph automorphisms provides a physical point of comparison for group-based shuffle models. Miyamoto and Shinagawa realize a uniformly chosen graph automorphism with pile-scramble shuffles [19]. Shinagawa and Miyamoto extend this construction to graphs and hypergraphs and apply it to cyclic group shuffles [27]. These physical protocols implement structured uniform shuffle groups exactly. This paper’s word approach offers a different tradeoff. A pile-scramble realization samples its group uniformly in one physical procedure, while a word shuffle repeats elementary moves and incurs a kernel-certified L_1 error of 2^{-40} . That line of work provides unobserved physical realizations for structured shuffles. I leave open whether the five $\text{PGL}(2, 7)$ letters admit

such realizations. The eight-card instance also has a larger deck vocabulary than two-symbol encodings. It stores a Boolean secret in the arrangement of eight pairwise distinct cards, whereas committed formats store one bit in two two-symbol cards [22]. Broader results show what physical shuffle models can support. A single uniform closed shuffle can compute any Boolean circuit [28]. A later compiler maps simple finite-runtime card protocols to private simultaneous-message protocols [29]. Because this paper uses an approximate word shuffle, the next comparison concerns the formal tools used to certify its mixing bound.

Machine-checked probability libraries provide results that can support mixing arguments. Karayel formalizes expander graphs, Cheeger’s inequality, and random-walk tail bounds in Isabelle/HOL [13]. Barthe, Grégoire, Hsu, and Strub develop a relational program logic for couplings and apply it to rapid-mixing examples [4]. Standard mathematical treatments connect spectral information with convergence of finite Markov chains [18]. The $\text{PGL}(2, 7)$ proof uses exact fiber counting instead. It counts the fibers at one finite word length and checks the resulting integer inequality. This method gives a concrete bound without requiring a general spectral development.

Proof-assistant frameworks for cryptography include CertiCrypt, CryptHOL, and SSProve [5,6,1]. They support program semantics, probabilistic reasoning, or modular cryptographic proofs at different levels. This paper proves information-theoretic bounds rather than game-based reductions. It therefore uses MathComp and infotheo for finite algebra, probability, independence, and entropy [30,2]. The earlier FORTE development provides the interpreter and trace semantics used here [31]. This paper extends those semantics with group-parametric card shuffles and coalition traces.

10 Conclusion and Future Work

I presented a group-parametric Rocq framework for card-based protocols and instantiated it with the action of $\text{PGL}(2, 7)$ on eight positions. Theorem A proves executed correctness, recovery parameters $(3, 7, 8)$, and perfect view and trace privacy for coalitions of at most three players under the uniform group distribution. The tuple $(3, 7, 8)$ gives the privacy threshold, recovery threshold, and total-player count. These security results assume a uniform Boolean prior, the fixed representative dealer, passive adversaries, and the dealer distributions in Section 2. The verifier learns the secret. Theorem 1 requires only a three-transitive action, a fixed encoding into decks of distinct cards, and the uniform group distribution. It therefore applies to any instance that supplies these three inputs.

For a finite shuffle implementation, Theorem B gives an L_1 distance of at most 2^{-40} after 200 generator letters. The same bound holds for every single-card endpoint and for the product with any Boolean secret prior. The approximate coalition-view and executed-trace bounds are 2^{-39} . The word-shuffle model therefore replaces the ideal uniform shuffle at a quantified, machine-checked cost.

The word-shuffle security results leave two direct extensions open. First, converting the L_1 bound into a conditional-entropy bound uses classical mathematics. Continuity of entropy in total variation is the Fannes inequality [11,9]. Only its formalization remains open here because the underlying probability library does not yet provide it. Second, the all-decks and shuffle-free dealers need word-shuffle mixing statements on their own sample spaces. Two broader directions would extend the framework. Connecting bounded-width branching programs with group-based card protocols through Barrington’s theorem [3,10] would make the workshop abstract’s NC^1 compilation proposal precise. Further work should also add active and compositional security by building on models for active card protocols [16]. A separate proof-completeness issue concerns the quantitative leakage for reveal sets of four through six cards. Section 6 settles these values at the current trust level. The orbit census and collision counts are kernel-checked, and a committed script evaluates the four values from these counts. Two kernel-level steps remain open. One is a proof of the mutual-information identity. The other is a proof of the orbit-invariance step that extends each value from its representative.

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